

SOFTWARE REQUIREMENTS SPECIFICATION

Car Auction Marketplace Platform — Canadian Market

Reverse-Auction Vehicle Selling Platform connecting Verified Sellers with Licensed Dealerships

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Document Control

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Approvals

Name	Role	Signature	Date
Faheem	Client / Product Owner		
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Distribution

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Table of Contents

Document Control.....	2
1. Introduction.....	4
1.1 Purpose.....	4
1.2 Scope.....	4
1.3 Definitions, Acronyms and Abbreviations.....	4
1.4 Canadian Regulatory Context.....	5
1.5 Intended Audience and Reading Guide.....	5
2. Overall Description.....	6
2.1 Product Perspective.....	6
2.2 User Classes and Characteristics.....	6
2.3 Operating Environment.....	6
2.4 Design and Implementation Constraints.....	6
2.5 Assumptions and Dependencies.....	6
3. Technology Stack.....	8
3.1 Stack Summary.....	8
3.2 Rationale.....	9
3.3 Environments.....	9
4. Functional Requirements.....	10
4.1 End User (Seller) Module.....	10
4.2 Dealership Module.....	11
4.3 Administrator Module.....	12
4.4 Cross-Cutting Requirements.....	12
5. Non-Functional Requirements.....	14
6. External Interface Requirements.....	15
7. System Diagrams.....	16
7.1 Use Case Diagram.....	16
7.2 Entity Relationship Diagram (ERD).....	18
7.3 System Architecture Diagram.....	21
7.4 Sequence Diagram — Listing Creation with VIN Decode.....	23
7.5 Sequence Diagram — Approval, Dealer Notification and Bidding.....	24
7.6 Sequence Diagram — Auction Close, Selection and Acceptance Window.....	25
7.7 Sequence Diagram — Dealer Subscription Lifecycle.....	27
8. Business Rules.....	28
9. Acceptance Criteria (Representative Samples).....	29
10. Risks and Mitigations.....	30
11. Delivery Phasing.....	31
12. Project Cost and Payment Schedule.....	32
13. Sign-Off.....	33

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for a web-based reverse-auction marketplace serving the Canadian automotive market. The platform enables private vehicle owners to list their cars for sale and allows subscribed, licensed dealerships to place competitive bids within a fixed 24-hour auction window. This document is the single source of truth for scope, behaviour, data design, and system architecture, and is intended for the client (Faheem), the Nexar Sol development team, QA engineers, and future maintainers.

1.2 Scope

The system serves three actor classes: End Users (vehicle sellers), Dealerships (buyers), and Administrators (platform operators). The core business flow is:

- A KYC-verified seller creates a vehicle listing, optionally auto-populated from a VIN decode API, including photos, damage declarations, and an asking price.
- All eligible subscribed dealerships within range are notified by email and push notification the moment the listing goes live.
- Dealerships place sealed bids during a 24-hour auction window.
- When the window closes, the seller reviews all bids and selects a dealership — either the highest bidder or the nearest one.
- Both parties are notified, and the seller has 24 hours to visit the chosen dealership to complete the transaction offline.

Out of scope for Version 1.0: in-platform payment processing for vehicle sales, vehicle transport/logistics, financing, and trade-in valuation tools.

1.3 Definitions, Acronyms and Abbreviations

Term	Definition
VIN	Vehicle Identification Number — a unique 17-character vehicle identifier (ISO 3779)
KYC	Know Your Customer — identity verification of end users before they may list vehicles
Auction Window	The fixed 24-hour period during which dealers may place bids on a listing
Acceptance Window	The fixed 24-hour period the seller has to visit the selected dealership
Sealed Bid	A bid whose amount is hidden from competing dealers and from the seller until the window closes
Match	The pairing of a listing with the winning dealership after seller selection
SRS	Software Requirements Specification (this document)
MVP	Minimum Viable Product — first shippable release
PIPEDA	Personal Information Protection and Electronic Documents Act (Canadian federal privacy law)
CASL	Canadian Anti-Spam Legislation

1.4 Canadian Regulatory Context

Because the platform operates in Canada and handles personal information, identity documents, and commercial electronic messages, the following regulatory obligations shape the requirements in this document:

- **PIPEDA** — federal privacy law governing collection, use, and disclosure of personal information; requires meaningful consent, limited retention, and breach notification to the Office of the Privacy Commissioner.
- **Quebec Law 25** — enhanced consent rules, a designated privacy officer, privacy impact assessments, and mandatory breach reporting for Quebec residents.
- **Bilingual requirement** — an English/French user interface is required for operation in Quebec; the platform is designed for full EN/FR parity.
- **Provincial dealer licensing** — dealership accounts must be verified against the relevant provincial regulator (OMVIC in Ontario, VSA in British Columbia, AMVIC in Alberta, and equivalents elsewhere).
- **CASL** — all commercial email and push notifications require express or implied consent and a functioning unsubscribe mechanism.

1.5 Intended Audience and Reading Guide

Sections 1–3 give business context and the agreed technology stack. Section 4 lists functional requirements per module. Sections 5–6 cover quality attributes and external interfaces. Section 7 contains the system diagrams (use case, ERD, architecture, and sequence diagrams). Sections 8–11 define business rules, acceptance criteria, risks, and delivery phasing.

2. Overall Description

2.1 Product Perspective

The platform is a standalone SaaS product. It integrates with external services for VIN decoding, identity verification (KYC), subscription payments, transactional email, push notifications, SMS, and geolocation. Vehicle sale payments themselves occur offline at the dealership and are outside the system boundary; the platform records only the outcome and final price.

2.2 User Classes and Characteristics

User Class	Description	Usage Pattern
End User (Seller)	Canadian resident, 18+, owner of the vehicle being listed. Low technical proficiency assumed; mobile-first experience.	Occasional — typically one listing at a time
Dealership	Licensed motor vehicle dealer with an active platform subscription. May have multiple staff accounts under one dealership entity.	Daily, high frequency — monitoring feed and bidding
Administrator	Nexar Sol / platform operations staff. Full oversight: moderation, verification, disputes, configuration, financial reporting.	Daily operational use

2.3 Operating Environment

- Responsive single-page web application supporting the latest two versions of Chrome, Safari, Firefox, and Edge.
- Mobile-first responsive design; installable PWA with web push (native mobile apps may be quoted as a future release).
- Server hosted on a Hostinger KVM 8 VPS (see Section 3); media stored in AWS S3, Canadian region.

2.4 Design and Implementation Constraints

- The auction window is fixed at exactly 24 hours from listing approval.
- The acceptance window is fixed at exactly 24 hours from dealer selection.
- Bids are sealed during the window and binding on the dealer, subject to physical inspection matching the declared condition.
- The platform never processes vehicle sale payments; only subscription billing is handled online.
- All monetary amounts are stored as integer cents in CAD; floating-point currency arithmetic is prohibited.

2.5 Assumptions and Dependencies

- A VIN decode API is available (NHTSA vPIC free tier as baseline; commercial providers such as Vehicle Databases, DataOne, or CarMD as upgrades).

- A KYC provider with Canadian document coverage is contracted (Trulioo, Persona, or Onfido).
- Stripe supports CAD recurring billing for dealership subscriptions.
- Dealership licence numbers can be verified against provincial registries (manually in MVP, API-assisted where available).

3. Technology Stack

The following stack has been agreed for this project. It favours a widely-adopted JavaScript ecosystem end to end, keeping the team, hiring, and long-term maintenance simple, while offloading heavy media storage to AWS.

3.1 Stack Summary

Layer	Technology	Details
Frontend	React 18	Single-page application; React Router, Redux Toolkit (or React Query) for state/data; component library (MUI or Tailwind CSS); i18next for EN/FR localization; PWA with service worker for web push.
Backend	Node.js (LTS) + Express	REST API in Node.js 20 LTS with Express; JWT authentication with refresh tokens; role-based access control; Joi/Zod request validation; BullMQ job queues for auction close, window expiry, and notification fan-out.
Database	SQL — MySQL 8	Relational schema per the ERD in Section 7.2; InnoDB with row-level locking for bid concurrency; spatial index for dealer-radius queries; Sequelize or Prisma ORM with versioned migrations. (PostgreSQL is an accepted drop-in alternative if preferred.)
Cache / Queue	Redis	Session store, rate limiting counters, auction countdown cache, and BullMQ backing store.
Version Control	GitHub	Private repository under the Nexar Sol organisation; GitFlow branching (main / develop / feature branches); protected main branch with mandatory pull-request review; GitHub Actions for CI (lint, test, build) and CD deployment to the server.
Server / Hosting	Hostinger KVM 8 VPS	8 vCPU, 32 GB RAM, 400 GB NVMe SSD, 32 TB bandwidth. Ubuntu 22.04 LTS; Nginx reverse proxy with SSL (Let's Encrypt); PM2 cluster mode for the Node.js API; MySQL and Redis co-located; UFW firewall + fail2ban hardening; automated nightly backups pushed off-server to AWS S3.
Image / Media Repository	AWS S3 (ca-central-1)	All vehicle photos and videos uploaded directly from the browser via pre-signed URLs (never through the API server); server-side encryption at rest; CloudFront CDN in front of the bucket for fast delivery; lifecycle rules to archive media of completed listings.
Email / Push / SMS	SendGrid + Firebase FCM + Twilio	Transactional and CASL-compliant notification delivery across channels.
Payments	Stripe (CAD)	Dealer subscription checkout, recurring billing, webhooks, dunning.
Monitoring	PM2 + UptimeRobot + Sentry	Process management, uptime alerting, and application error tracking.

3.2 Rationale

- **React + Node.js:** one language across the stack shortens development time, and both have the largest talent pools for future maintenance.
- **SQL (MySQL):** the domain is strongly relational — users, vehicles, listings, bids, matches, subscriptions — and requires ACID transactions for race-safe auction close and bid revision. A document database would add risk without benefit here.
- **Hostinger KVM 8:** generous fixed-cost resources (8 vCPU / 32 GB RAM) comfortably support the MVP and early growth as a monolith, with full root control. When traffic outgrows a single VPS, the stateless API design allows horizontal scaling behind a load balancer without re-architecture.
- **AWS S3 for images:** vehicle listings are media-heavy (8–30 photos each). Keeping media off the VPS protects its disk and bandwidth, gives 11-nines durability, and the ca-central-1 region keeps data on Canadian soil for privacy compliance.
- **GitHub:** industry-standard collaboration, protected branches, code review, and free CI/CD via GitHub Actions with SSH deployment to the Hostinger VPS.

3.3 Environments

Environment	Purpose	Location
Development	Local developer machines; Docker Compose for MySQL/Redis parity	Developer workstations
Staging	Client review and QA; mirrors production configuration	Hostinger VPS (isolated instance or subdomain)
Production	Live platform	Hostinger KVM 8 + AWS S3/CloudFront

4. Functional Requirements

4.1 End User (Seller) Module

ID	Requirement	Description
FR-EU-01	Registration	Register via email/password or Google OAuth. Required: full name, email, mobile number, postal code, province. Email and SMS verification are mandatory before login is enabled.
FR-EU-02	KYC Verification	KYC must be completed before the first listing. The user uploads a government-issued photo ID (driver's licence or passport) plus a liveness selfie; the system submits to the KYC provider and stores only the result, provider reference, and status — not raw document images beyond the provider's retention. States: PENDING, VERIFIED, REJECTED, EXPIRED. Admin may override on appeal.
FR-EU-03	VIN Decode	Accept a 17-character VIN, validate its check digit (ISO 3779), and call the VIN API to auto-populate year, make, model, trim, body style, engine, transmission, drivetrain, and fuel type. On API failure the form falls back to manual entry. Decoded fields remain editable, with an "edited" flag retained for dealer transparency.
FR-EU-04	Listing Creation	Capture: VIN, year/make/model/trim, mileage (km), exterior/interior colour, transmission, drivetrain, number of keys, accident history, lien status and lienholder, service history notes, tire condition, damage records, and the asking price in CAD.
FR-EU-05	Media Upload	Minimum 8 photos (front ¾, rear ¾, both sides, interior front, interior rear, odometer, engine bay), maximum 30; up to 10 MB each. Images are compressed and EXIF-stripped, then uploaded directly to AWS S3 via pre-signed URLs. Optional 60-second walkaround video.
FR-EU-06	Damage Declaration	Tag damage on an interactive vehicle body diagram by zone, each with severity (Minor / Moderate / Severe), a description, and at least one photo.
FR-EU-07	Submission & Moderation	On submission the listing enters PENDING_REVIEW. An administrator approves or rejects it (with reason) within the agreed SLA. On approval the status becomes ACTIVE and the 24-hour auction window opens.
FR-EU-08	Sealed Bid Visibility	During the auction window the seller sees only the count of bids received — never individual amounts or dealer identities. Full results are revealed only when the window closes.
FR-EU-09	Results & Dealer Selection	At window close the seller sees all bids ranked by amount, with each dealership's name, rating, distance from the seller's postal code, and any bid conditions. The seller may sort by highest bid or nearest distance, and selects exactly one dealership.
FR-EU-10	Acceptance Window	On selection, both parties receive email and push notifications containing the dealership address, contact details, business hours, a required-documents checklist, and a countdown expiring in 24 hours. If the window expires without a recorded visit, the match is released and the seller may pick the next-ranked dealer or relist.

ID	Requirement	Description
FR-EU-11	Dashboard	Show active listings with live bid counts and countdowns, closed auctions with outcomes, pending acceptance windows, historical sales, and a notification centre.
FR-EU-12	Reports	Generate and export (PDF / CSV): listing history, bid summary per listing, and transaction records.
FR-EU-13	Relisting	Permit relisting when no bids were received, all bids were declined, or the acceptance window expired; a configurable cooldown (default 24 hours) applies.

4.2 Dealership Module

ID	Requirement	Description
FR-DL-01	Registration	Capture legal business name, operating name, business number (BN), provincial dealer licence number, licensing body, dealership address, primary contact, and service radius in kilometres.
FR-DL-02	Dealer Verification	An administrator verifies the dealer licence against the relevant provincial registry before activation. States: PENDING, VERIFIED, SUSPENDED, REVOKED.
FR-DL-03	Subscription & Fees	Tiered plans (Basic / Pro / Enterprise) differentiated by monthly bid quota, notification radius, and analytics depth; monthly and annual billing in CAD via Stripe, including proration, upgrades, downgrades, cancellation, and dunning on failed payment. Bidding is blocked while the subscription is EXPIRED or PAST_DUE.
FR-DL-04	Multi-User Accounts	A dealership supports multiple staff accounts with roles: DEALER_ADMIN (billing, user management, bidding) and DEALER_AGENT (bidding only).
FR-DL-05	New-Listing Notification	On listing approval, dispatch email and push notification to every verified dealership with an active subscription whose service radius contains the listing location. The notification includes a vehicle summary, thumbnail, and deep link. Dispatch SLA: within 60 seconds of approval.
FR-DL-06	Auction Feed	A filterable feed of open auctions (make, model, year range, mileage, distance, time remaining, damage severity) with full vehicle detail, photo gallery, damage report, and live countdown.
FR-DL-07	Bid Placement	Submit a bid amount in CAD with optional conditions and notes. Validation: subscription active, quota available, amount greater than zero , window open. Bids are sealed from competing dealers. A dealer may revise their own bid upward any number of times before close; downward revision is prohibited; withdrawal requires admin approval.
FR-DL-08	Bid Lifecycle	SUBMITTED → ACTIVE → (WON LOST EXPIRED WITHDRAWN).
FR-DL-09	Dealer Dashboard	Active bids with countdowns, closed bids with outcomes, won auctions with pending appointments, win rate, average winning bid,

ID	Requirement	Description
		remaining bid quota, and subscription status.
FR-DL-10	Appointment Management	On winning, the dealer receives the seller's contact details and the appointment countdown, and records the outcome: COMPLETED (with final price), NO_SHOW, or DECLINED_AFTER_INSPECTION with a reason.
FR-DL-11	Reports	Export bid history, win/loss analysis, spend by period, acquisition cost per vehicle, and subscription invoices.

4.3 Administrator Module

ID	Requirement	Description
FR-AD-01	User Management	View, search, filter, suspend, reactivate, and delete end-user accounts; review KYC submissions; process rejection appeals.
FR-AD-02	Dealer Management	Approve or reject dealership registrations; verify licences; suspend for policy violations; view dealer activity and history.
FR-AD-03	Listing Moderation	Review pending listings; approve, reject with reason, or request edits; flag suspected fraud for investigation.
FR-AD-04	Auction Oversight	View all auctions in every state; manually extend, cancel, or void an auction with a mandatory logged reason.
FR-AD-05	Subscription Management	Define plans and pricing; view all subscriptions; issue refunds, credits, and complimentary access; manage dunning.
FR-AD-06	Dispute Resolution	Case-management queue for seller/dealer disputes with categories, status tracking, assignment, and resolution notes.
FR-AD-07	Platform Reports	Gross merchandise value, auction volume, fill rate, average bids per listing, average sale price by make/model/province, dealer engagement, subscription MRR/ARR and churn, KYC pass rate, and a geographic heat map.
FR-AD-08	Configuration	Adjust auction duration, acceptance window, default notification radius, minimum photo count, and relisting cooldown without code changes.
FR-AD-09	Audit Log	Immutable log of every admin action recording actor, timestamp, entity, and before/after values.

4.4 Cross-Cutting Requirements

ID	Requirement	Description
FR-X-01	Notifications	Email (SendGrid), push (Firebase FCM), and optional SMS (Twilio) with per-user channel preferences. All commercial messages are CASL-compliant with a working unsubscribe.
FR-X-02	Scheduler	A reliable job scheduler (BullMQ on Redis) drives auction close,

ID	Requirement	Description
		acceptance-window expiry, subscription renewal, and reminder dispatch. Jobs are idempotent and survive server restarts; a reconciliation sweep catches any missed executions.
FR-X-03	Geolocation	Resolve postal codes to coordinates and compute the distance between seller and dealership for radius targeting and result ranking.
FR-X-04	Bilingual Support	Full English/French parity across UI, emails, and generated documents.
FR-X-05	Search & Filter	Indexed search across listings for the dealer feed with faceted filters.

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Page load under 2 seconds (95th percentile); API responses under 500 ms (95th percentile).
NFR-02	Performance	Bid submission processed in under 200 ms.
NFR-03	Scalability	Support 5,000 concurrent users and 1,000 simultaneous open auctions on the KVM 8 VPS, with a documented horizontal-scaling path beyond that.
NFR-04	Availability	99.9% platform uptime; auction close and bid intake treated as critical paths.
NFR-05	Security	TLS 1.3 in transit; AES-256 encryption at rest for the database and S3; bcrypt/argon2 password hashing.
NFR-06	Security	Multi-factor authentication mandatory for administrator and dealer-admin roles.
NFR-07	Security	Rate limiting on authentication and bid endpoints; OWASP Top 10 mitigations; UFW firewall and fail2ban on the VPS.
NFR-08	Privacy	PIPEDA and Quebec Law 25 compliance; explicit consent capture; data-subject access and deletion honoured within 30 days.
NFR-09	Data Residency	Media stored in AWS ca-central-1 (Canada); database backups encrypted and retained on Canadian-region storage; PIPEDA breach notification procedure documented.
NFR-10	Reliability	Automated database backups every 6 hours pushed to S3; recovery point objective 1 hour; recovery time objective 4 hours.
NFR-11	Auditability	Immutable audit trail retained for 7 years.
NFR-12	Usability	WCAG 2.1 AA accessibility compliance.
NFR-13	Localization	Full EN/FR parity; CAD currency; metric units (kilometres).
NFR-14	Data Integrity	All monetary amounts stored as integer cents; no floating-point currency arithmetic.
NFR-15	Concurrency	Optimistic locking on bid revision; transactional, idempotent, race-safe auction close.
NFR-16	Maintainability	GitHub pull-request review mandatory; minimum 70% unit-test coverage on the auction engine and billing logic; CI must pass before merge to main.

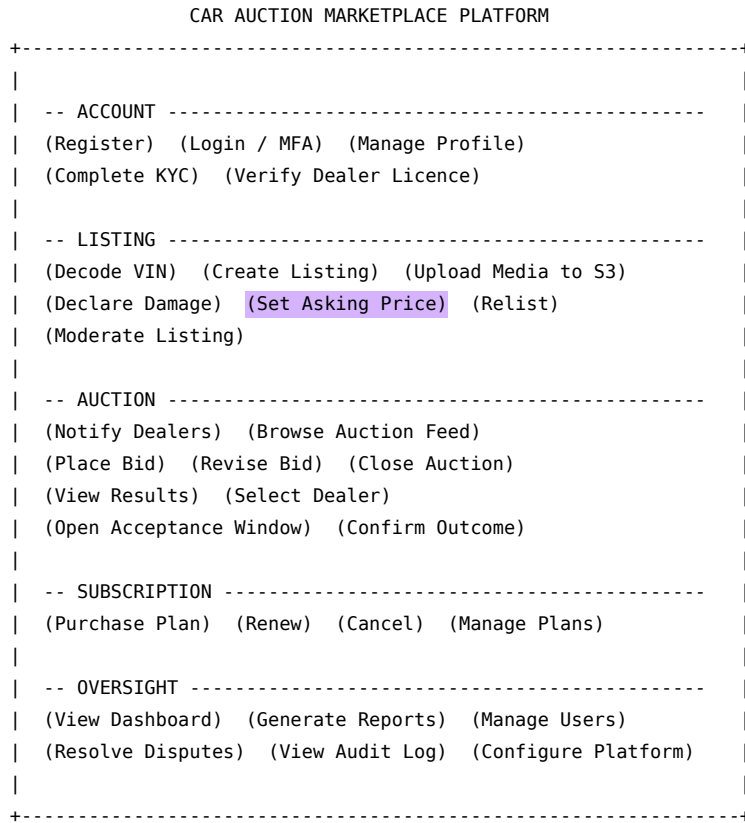
6. External Interface Requirements

Interface	Provider	Purpose
VIN Decode API	NHTSA vPIC (free) → Vehicle Databases / DataOne / CarMD (commercial)	Auto-populate vehicle specifications from the VIN
Vehicle History	CARFAX Canada / AutoCheck (future release)	Accident and lien history records
KYC Provider	Trulioo / Persona / Onfido	Government-ID and liveness verification of sellers
Payments	Stripe (CAD, recurring)	Dealer subscription checkout, billing, webhooks, dunning
Email	SendGrid	Transactional and notification email
Push	Firebase Cloud Messaging	Web and mobile push notifications
SMS	Twilio	OTP verification and critical alerts
Geocoding	Google Maps Geocoding / Canada Post AddressComplete	Postal-code to coordinate resolution and distances
Media Storage	AWS S3 (ca-central-1) + CloudFront	Vehicle photo/video repository and CDN delivery
Version Control / CI	GitHub + GitHub Actions	Source control, code review, automated test and deploy

7. System Diagrams

The following diagrams are rendered in monospace notation for portability. Editable Figma / draw.io versions can be produced on request during the design phase.

7.1 Use Case Diagram



ACTORS AND ASSOCIATIONS

[End User / Seller] --> Register, Login/MFA, Manage Profile,
Complete KYC, Decode VIN, Create Listing, Upload Media,
Declare Damage, Set Asking Price, Relist, View Results,
Select Dealer, Confirm Outcome, View Dashboard, Reports

[Dealership] --> Register, Login/MFA, Verify Dealer Licence,
Purchase Plan, Renew, Cancel, Browse Auction Feed,
Place Bid, Revise Bid, Confirm Outcome, Dashboard, Reports

[Administrator] --> Manage Users, Moderate Listing,
Verify Dealer Licence, Manage Plans, Resolve Disputes,
View Audit Log, Configure Platform, Reports,
Close Auction (manual override)

[System Scheduler] --> Notify Dealers, Close Auction,
Open Acceptance Window, Expire Window, Renew Subscription


```
[VIN API]    <-- <<include>> Decode VIN
[KYC API]    <-- <<include>> Complete KYC
[Stripe]    <-- <<include>> Purchase Plan, Renew
[Email/Push] <-- <<include>> Notify Dealers
```

RELATIONSHIPS

```
* Create Listing <<include>> Decode VIN
* Create Listing <<include>> Upload Media to S3
* Create Listing <<extend>> Declare Damage [if damage present]
* Place Bid      <<include>> Validate Subscription
* Select Dealer  <<include>> Open Acceptance Window
* Close Auction  <<extend>> Relist [if zero bids]
```

Figure 7.1 — Use case diagram: actors, use cases, and include/extend relationships

7.2 Entity Relationship Diagram (ERD)

+-----+		+-----+	
	USER		KYC_VERIFICATION
+-----+ 1		1 +-----+	
	PK user_id		PK kyc_id
	email (UQ)		FK user_id (UQ)
	password_hash		provider
	phone		provider_ref
	first_name		status
	last_name		id_doc_type
	postal_code		verified_at
	province		expires_at
	latitude		reject_reason
	longitude	+-----+	
	role	+-----+	
	language_pref	+-----+	
	mfa_enabled	1	N NOTIFICATION_PREF
	status	+-----+	
	created_at		PK pref_id
+-----+			FK user_id
	1		channel
			event_type
	N		enabled
+-----+		+-----+	
	VEHICLE	+-----+	
+-----+		+-----+	
	PK vehicle_id	1	N VEHICLE_PHOTO
	FK owner_user_id	+-----+	
	vin (UQ)		PK photo_id
	year, make		FK vehicle_id
	model, trim		s3_key
	body_style		photo_type
	mileage_km		sort_order
	transmission		uploaded_at
	drivetrain	+-----+	
	fuel_type	+-----+	
	ext_colour	+-----+	
	int_colour	1	N DAMAGE_RECORD
	accident_history	+-----+	
	lien_status		PK damage_id
	lienholder		FK vehicle_id
	keys_count		zone
	vin_decoded		severity
	manual_edits		description
	created_at		photo_s3_key
+-----+		+-----+	
	1		
	1		
+-----+		+-----+	
	LISTING	status: DRAFT PENDING_REVIEW	
+-----+		ACTIVE CLOSED MATCHED	
	PK listing_id	COMPLETED EXPIRED	
	FK vehicle_id (UQ)	CANCELLED RELISTED	

FK seller_user_id			
asking_price_cts			
description			
status			
approved_by			
approved_at			
published_at			
auction_ends_at			
relist_count			
parent_listing_id			
+-----+			
1		+-----+	
		DEALERSHIP	
N		+-----+	
+-----+	N	1	PK dealership_id
BID			legal_name
+-----+			operating_name
PK bid_id			business_number
FK listing_id			licence_number
FK dealership_id			licensing_body
FK placed_by_user			address, city
amount_cents			province
conditions			postal_code
notes			latitude
status			longitude
revision_count			service_radius_km
placed_at			rating
updated_at			status
UQ(listing,dealer)			verified_at
+-----+			+-----+
1		1	1
1		N	N
+-----+			+-----+
MATCH		DEALERSHIP_USER	SUBSCRIPTION
+-----+			+-----+
PK match_id		PK dl_user_id	PK sub_id
FK listing_id (UQ)		FK dealership_id	FK dealership
FK winning_bid_id		FK user_id	FK plan_id
FK dealership_id		dealer_role	status
FK seller_user_id		invited_by	period_start
selection_reason		joined_at	period_end
accept_expires_at		+-----+	bids_used
status			auto_renew
outcome			gateway_ref
outcome_notes			cancelled_at
final_price_cts			+-----+
created_at			N
+-----+			1
			+-----+
+-----+			PLAN
NOTIFICATION		PAYMENT	+-----+
+-----+			PK plan_id
PK notification_id		PK payment_id	name
FK recipient_user_id		FK sub_id	price_cents

FK listing_id (NULL)		amount_cents		interval	
channel		currency=CAD		bid_quota	
event_type		status		radius_km	
subject, body		gateway_txn		features	
status		invoice_url		is_active	
sent_at, read_at		paid_at		+-----+	
+-----+		refunded_at			
+-----+					
DISPUTE		AUDIT_LOG			
+-----+					
PK dispute_id		PK audit_id			
FK match_id		FK actor_user_id			
FK raised_by_user		entity_type			
FK assigned_admin		entity_id			
category		action			
description		before_json			
status		after_json			
resolution		ip_address			
resolved_at		created_at			
+-----+					

CARDINALITY SUMMARY

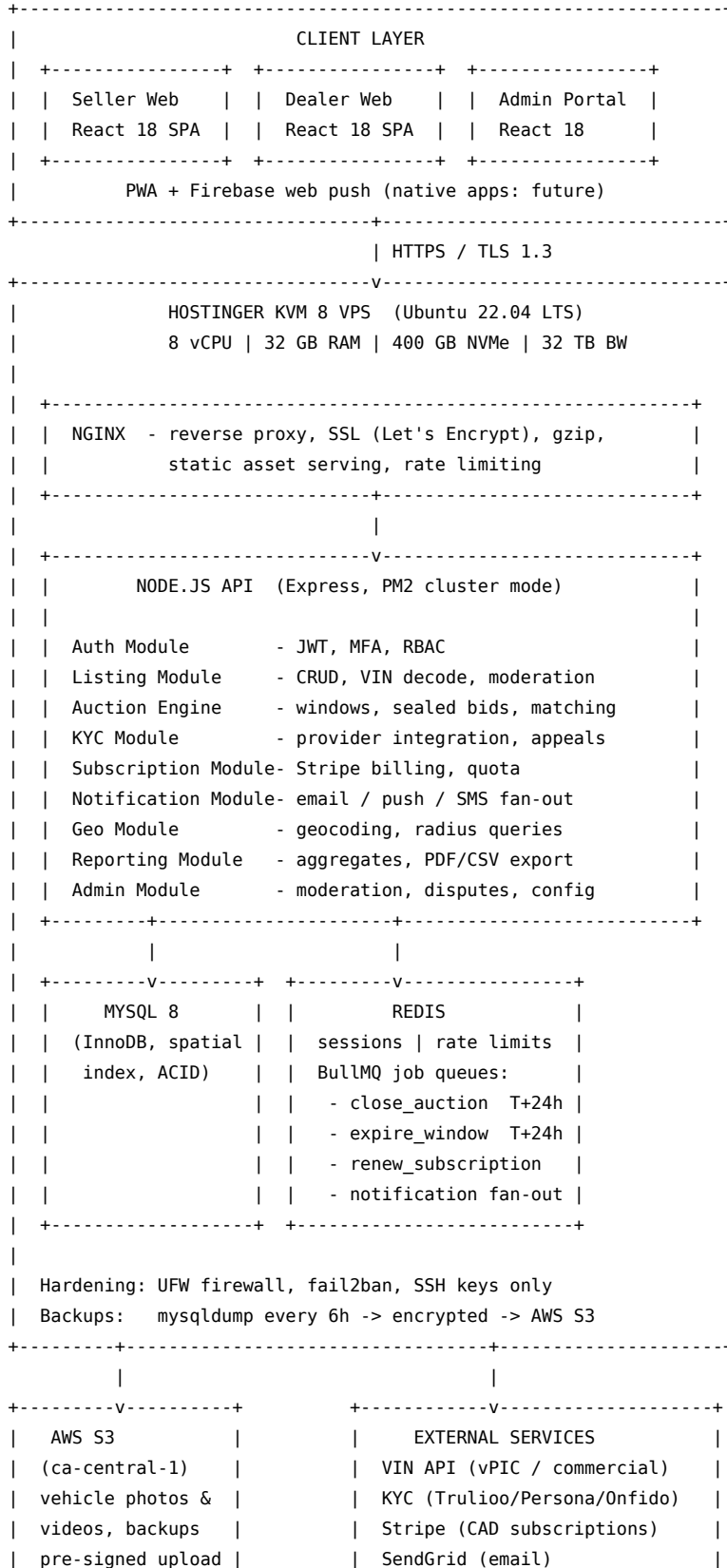
USER	1 -- 0..1	KYC_VERIFICATION
USER	1 -- 0..N	VEHICLE
VEHICLE	1 -- 1..N	VEHICLE_PHOTO
VEHICLE	1 -- 0..N	DAMAGE_RECORD
VEHICLE	1 -- 0..1	LISTING
LISTING	1 -- 0..N	BID
LISTING	1 -- 0..1	MATCH
DEALERSHIP	1 -- 1..N	DEALERSHIP_USER
DEALERSHIP	1 -- 0..N	BID
DEALERSHIP	1 -- 0..N	SUBSCRIPTION
PLAN	1 -- 0..N	SUBSCRIPTION
SUBSCRIPTION	1 -- 0..N	PAYMENT
BID	1 -- 0..1	MATCH
MATCH	1 -- 0..N	DISPUTE

KEY CONSTRAINTS

- * UNIQUE (listing_id, dealership_id) on BID
 - one active bid per dealer per listing
- * CHECK amount_cents > 0
- * auction_ends_at = published_at + 24 hours
- * accept_expires_at = match.created_at + 24 hours
- * Index BID(listing_id, amount_cents DESC)
- * Spatial index on DEALERSHIP(latitude, longitude)

Figure 7.2 — Entity relationship diagram (logical model, MySQL 8)

7.3 System Architecture Diagram



7.4 Sequence Diagram — Listing Creation with VIN Decode

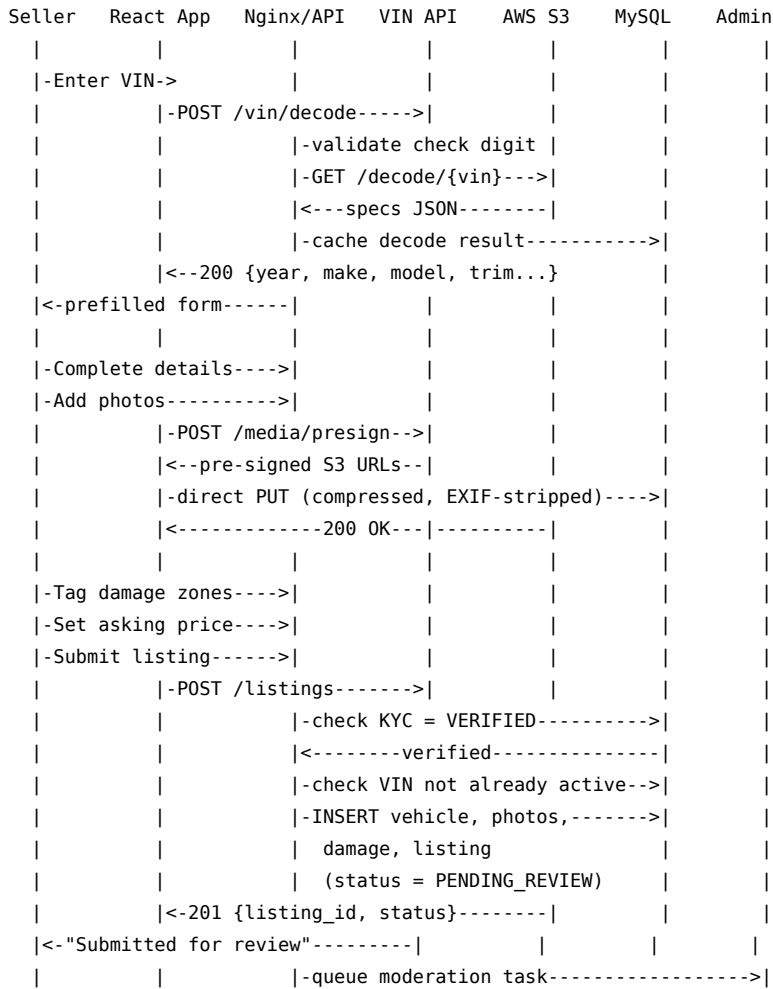


Figure 7.4 — Seller creates a listing; VIN decode and direct-to-S3 media upload

7.6 Sequence Diagram — Auction Close, Selection and Acceptance Window

```

sequenceDiagram
    participant BullMQ
    participant Auction Eng
    participant MySQL
    participant Notif
    participant Seller
    participant Dealer(won)
    participant Dealer(lost)

    BullMQ->>Auction Eng: |-fire close_auction(listing_id) at T+24h|
    Auction Eng->>MySQL: |----->|
    MySQL->>Notif: |-BEGIN TRANSACTION->|
    MySQL->>Notif: |-SELECT listing FOR UPDATE->|
    MySQL->>Notif: |-guard: status = ACTIVE (idempotent)|
    MySQL->>Notif: |-UPDATE listing status=CLOSED->|
    MySQL->>Notif: |-SELECT bids ORDER BY amount DESC->|
    MySQL->>Notif: |-compute dealer distance ->|
    MySQL->>Notif: |-COMMIT--->|
    MySQL->>Notif: |-emit auction.closed->|
    MySQL->>Notif: |-email + push----->|
    MySQL->>Notif: "Auction closed - N bids received"
    MySQL->>Notif: |-View results->|
    MySQL->>Notif: |<-ranked: dealer name,|
    MySQL->>Notif: | amount, distance km,|
    MySQL->>Notif: | rating, conditions|
    MySQL->>Notif: |-Select dealer----->|
    MySQL->>Notif: | POST /matches|
    MySQL->>Notif: | {bid_id, reason:|
    MySQL->>Notif: | HIGHEST or NEAREST}|
    MySQL->>Notif: |-BEGIN TRANSACTION->|
    MySQL->>Notif: |-guard: no existing match (UNIQUE listing_id)|
    MySQL->>Notif: |-INSERT match,|
    MySQL->>Notif: | accept_expires_at = now + 24h->|
    MySQL->>Notif: |-UPDATE winning bid = WON->|
    MySQL->>Notif: |-UPDATE other bids = LOST->|
    MySQL->>Notif: |-UPDATE listing = MATCHED->|
    MySQL->>Notif: |-COMMIT--->|
    MySQL->>Notif: |<-enqueue expire_window(match_id) delay=24h|
    MySQL->>Notif: |-emit match.created->|
    MySQL->>Notif: |-email + push----->|
    MySQL->>Notif: "You have 24 hours to visit [Dealer]:|
    MySQL->>Notif: | address, hours, documents checklist"|
    MySQL->>Notif: |-email + push----->|
    MySQL->>Notif: "You won - seller contact + details"|
    MySQL->>Notif: |-email (batch)----->|
    MySQL->>Notif: "Auction closed - bid not selected"|
    MySQL->>Notif: "=== OFFLINE: dealership inspection ==="|
    MySQL->>Notif: |-Confirm--->|
    MySQL->>Notif: | POST /matches/{id}/outcome|
    MySQL->>Notif: | {outcome: COMPLETED,|
    MySQL->>Notif: | final_price_cents: 2150000}|
    MySQL->>Notif: |-UPDATE match outcome,|
    MySQL->>Notif: | listing = COMPLETED----->|
    MySQL->>Notif: |-cancel expire_window job->|
    MySQL->>Notif: |-confirmation email->|

```

```

|           |           |           |           |           |
| === ALTERNATE: acceptance window expires, no outcome === |
|           |           |           |           |           |
|-fire expire_window(match_id) at T+24h |           |
|----->|           |           |           |           |
|           |-guard: match.outcome IS NULL->           |
|           |-UPDATE match = EXPIRED--->|           |
|           |-UPDATE listing = CLOSED-->|           |
|           |           |-notify both parties->           |
|           |           | "Window expired - select next dealer |
|           |           | or relist" |           |

```

Figure 7.6 — Scheduled close, seller selection, 24-hour acceptance window, and expiry path

8. Business Rules

ID	Rule
BR-01	KYC status must be VERIFIED before a user may publish a listing.
BR-02	Only one active listing per VIN may exist at any time.
BR-03	The auction window is exactly 24 hours from the approval timestamp.
BR-04	Bids are sealed; the seller sees only a bid count until the window closes.
BR-05	One active bid per dealership per listing; revisions may only increase the amount.
BR-06	A dealer must hold an ACTIVE subscription with available quota to bid.
BR-07	The seller may select any bid — the highest, the nearest dealership, or otherwise at their discretion.
BR-08	The acceptance window is exactly 24 hours from the moment of selection.
BR-09	On acceptance-window expiry, the seller may select the next-ranked bid or relist.
BR-10	Bids are binding offers, contingent on the physical inspection matching the declared condition.
BR-11	Undeclared damage discovered at inspection permits dealer withdrawal without penalty.
BR-12	Listings that receive zero bids may be relisted after the configured cooldown.
BR-13	A dealer licence must be verified against the provincial registry before account activation.
BR-14	Suspended dealers cannot bid, and their existing active bids are voided.



9. Acceptance Criteria (Representative Samples)

Full Gherkin-style acceptance suites will be authored per user story during sprint planning. The samples below establish the expected format and rigour.

AC-01 — VIN Decode

Given a valid 17-character VIN, **when** the seller submits it, **then** the system populates year, make, model, and trim within 3 seconds, or presents manual entry if the VIN API fails.

AC-02 — Auction Close

Given an active auction with 5 bids, **when** 24 hours elapse from publication, **then** the listing status changes to CLOSED, the seller is notified within 60 seconds, and all bid amounts become visible to the seller.

AC-03 — Bid Blocking

Given a dealer whose subscription is PAST_DUE, **when** they attempt to place a bid, **then** the request is rejected with HTTP 402 and a subscription-renewal prompt is displayed.

AC-04 — Sealed Bids

Given an active auction, **when** the seller views the listing, **then** only the bid count is displayed and no bid amount appears anywhere in the API response payload.

AC-05 — Acceptance Expiry

Given a match created 24 hours ago with no recorded outcome, **when** the expiry job fires, **then** the match status becomes EXPIRED, both parties are notified, and the seller may select the next-ranked bid.

AC-06 — Dealer Notification

Given an approved listing and 12 eligible dealerships in radius, **when** the listing goes ACTIVE, **then** all 12 dealerships receive email and push notification within 60 seconds, and the dispatch is logged.

10. Risks and Mitigations

Risk	Impact	Mitigation
Fraudulent listings (e.g., stolen vehicles)	High	KYC verification, VIN uniqueness check, admin moderation, lien declaration, and optional future CARFAX integration.
Dealer collusion suppressing bid prices	High	Sealed bids plus anomaly detection on bidding patterns across dealerships.
Low dealer density in rural regions	Medium	Configurable radius expansion and a fallback province-wide notification tier.
VIN API downtime	Medium	Multi-provider fallback, response caching, and graceful manual-entry degradation.
Off-platform circumvention after match	High	Seller contact details released only at match; mandatory outcome confirmation; dealer rating penalties for unreported completions.
Scheduler failure missing an auction close	High	Durable BullMQ jobs with retries plus a reconciliation sweep that closes any overdue auctions.
Single-VPS resource exhaustion	Medium	PM2 clustering across 8 vCPUs, Redis caching, S3 media offload, monitoring alerts, and a documented path to a load-balanced multi-node topology.
Quebec Law 25 non-compliance	High	Designated privacy officer, consent management, and French-language parity from launch.
